CPU and RAM

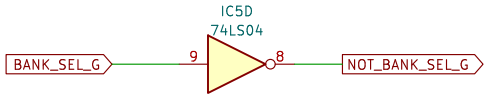
Arquivo: cpu_ram.kicad_sch

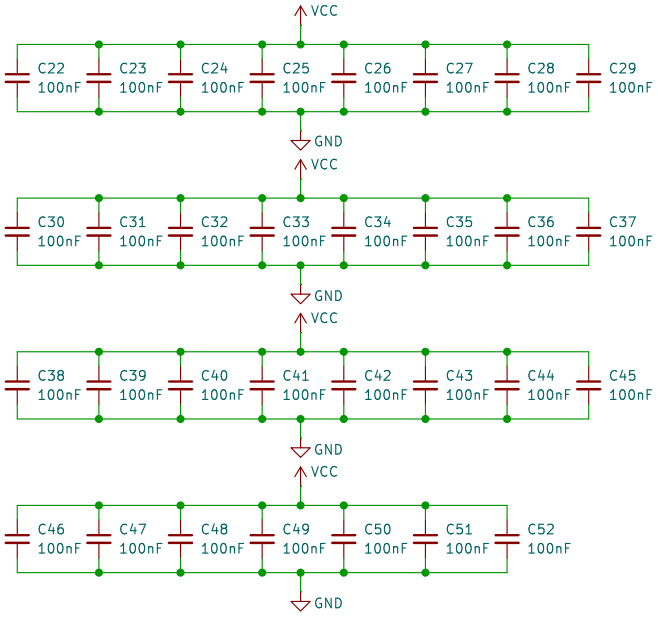
Expansion and Power Connectors

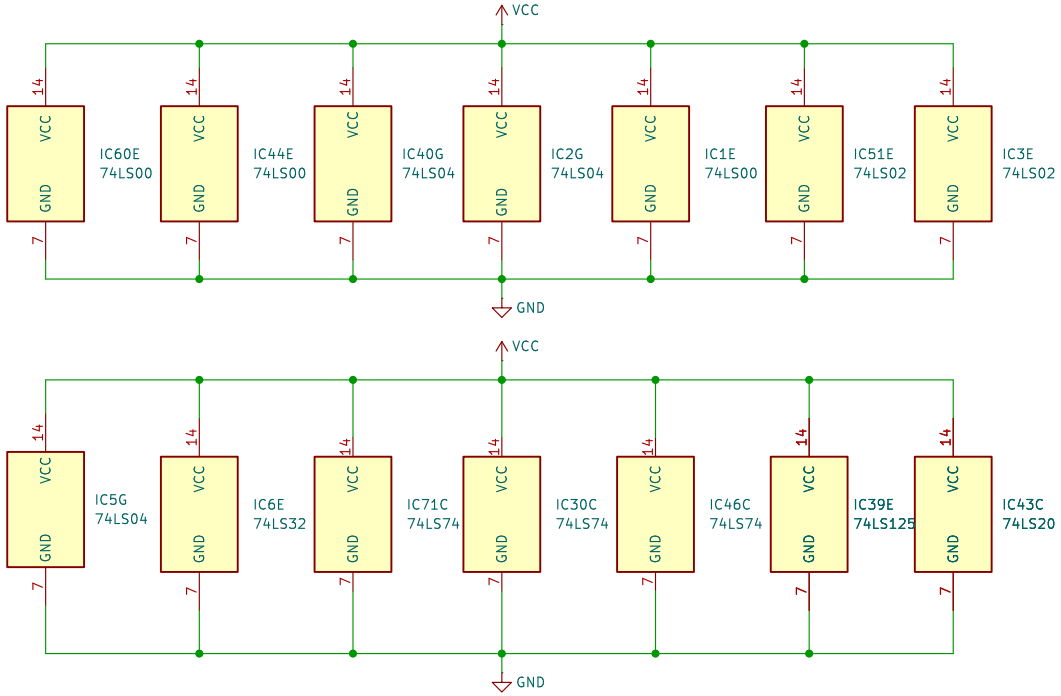
Arquivo: exp_conn.kicad_sch

IO and Video

Arquivo: io.kicad_sch







Sheet: /		
File: NE_Delta.kicad_sch		
Title: N.E. DELTA		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 1/4

While the original BOM specifies a 74LS04 for the oscillator, I had much better stability using either an 74ALS04 or a plain 7404.

Note that the S/PIN10 on IC30 is not mentioned in the original schematic.

In the original schematic PIN38 is connected to both GND and a pullup...

Sheet: /CPU and RAM/	
File: cpu_ram.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 9.0.2	**Rev:** Id: 2/4

Note that in the original schematic
B_SYNC is connected to CONN.7 only.
And yet, the PCB connects it through
all the other bus connectors.

SYNC
ABTLTX
P2
P1
CONN.7
CONN.6
CONN.5
CONN.4
CONN.3
CONN.2
CONN.1
VCC
GND
+12V
-5V
-12V
I/O7
I/O6
I/O5
I/O4
I/O3
I/O2
I/O1
B_SYNC
I/O stb
RDY
DMA
R/W
RES
IRQ
NMI
INM
7M
Q3
Φ1
µPSINC
Φ0

IC4
74LS138
IC3C
74LS02
SLCx
SCxxx
74LS00
IC1A
IC2E
74LS04
A11
IC5C
74LS04
SLC3
74LS32
IC6A
ROM C
A[0..15]
I/O1
I/O2
I/O3

IC7
74LS138
DEV0
DEV1
DEV2
DEV3
DEV4
DEV5
DEV6
DEV7

D[0..7]

CONN.8
POWER
C1 100µF
C2 100nF
C3 100nF
C4 100nF
C5 100nF
C6 100µF

Sheet: /Expansion and Power Connectors/
File: exp_conn.kicad_sch
Title:
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Date:
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IC49 74LS165
IC50 74LS374
IC51C 74LS02
IC20 74LS04
IC52 74LS374
IC53 74LS157
IC54 GAL16V8
CONN.9 Video Connector
CONN.11 Game Connector (DE9)
CONN.12 Game Connector (DIP16)
IC55 2732
IC56 74LS153
IC57 GAL16V8
IC58 74LS259
IC59 74LS283
IC61 74LS157
IC62 6545
IC63 74LS251
IC64 74LS259
IC65 74LS251
IC66 74LS251
IC67 NE556
IC68 NE556
IC69 74LS138
IC70 µA741
IC71A 74LS74
IC71B 74LS74
IC72 74LS244
IC73 NE556
IC30A 74LS74
IC51D 74LS02
IC40A 74LS04
IC40B 74LS04
IC40D 74LS04
IC40F 74LS04
IC60A 74LS00
IC60B 74LS00
IC60D 74LS00
IC60C 74LS00
IC10 74LS00
IC2B 74LS04
IC2C 74LS04
IC1B 74LS00
IC5B 74LS04
IC5E 74LS04
IC6C 74LS32
IC68 74LS32
P3
P4
P5
P6
P7
P8
R11 1K
R12 47K
R13 1K
R14 1K
R15 560
R16 560
R17 560
R18 560
R19 12K
R20 1M
R21 12K
R22 12K
R23 12K
R24 12K
R25 47K
R26 100
R27 27
R28 1K
R29 10K
R30 2.2M
R31 12K
R32 3.3M
C8 100µF
C9 22nF
C10 22nF
C11 22nF
C12 22nF
C13 22nF
C14 22nF
C15 22nF
C17 100nF
C18 100nF
C19 100nF
C20 100nF
C21 10µF
D1 1N4148
D2 1N4148
TR1 BC517
TR2 2N2222A
J1 Tape IN
J2 Tape OUT
LS1 Speaker
CONN.10 Keyboard Connector (DIP16)
SRLD
SRCK
VD[0..7]
S4080
VA[0..15]
CS6545
Φ1
Φ0
RES
D[0..7]
RDram
WRram
SLCx
ALTZP
SLC3
A[0..15]
R/W
READ_EN_G
NOT_BANK_SEL_G
SYNC
I/O0
ABTLTX
HIR
S1B0
TXT
PAGE2
VCC
GND
+12V
-12V
-5V

Sheet: /IO and Video/
File: io.kicad_sch

Title:

Size: A2 | Date: | Rev:
KiCad E.D.A. 9.0.2 | Id: 4/4